$$\widehat{w}_i = \frac{w_i}{\sum_{j=1}^{N_c} w_j}, \quad u \sim U[0,1],$$

$$t_{\text{new}} = \text{CDF}^{-1}(u),$$

$$N_c = 64, \quad N_f = 128,$$

$$\text{queries per ray} = N_c + (N_c + N_f) = 256$$